

Human Review: $c_0 \subset \ell_\infty$ closed-bounded radii

Original automatic proof: `solution_packet.pdf`

Checked date: 10 June 2026

Status: Manually checked.

Verdict: Correct, but not a full solution to the stated characterisation problem.

Human Comments

The argument is correct. The example $c_0 \subset \ell_\infty$ shows that equality of radii for finite subsets does not extend automatically to all closed bounded subsets.

However, this should not be read as a real solution to the broader problem as stated, since the problem asks for characterisations of when equality holds. The packet instead gives a natural obstruction: even when the relevant finite-dimensional or finite-subset conditions hold, they may fail to control the behaviour of closed bounded subsets.

In the example, every finite subset $F \subset c_0$ has

$$\text{rad}_{c_0}(F) = \text{rad}_{\ell_\infty}(F),$$

because a centre in ℓ_∞ can be truncated to an element of c_0 without losing the radius bound on finitely many vectors. But for

$$A = \{e_n : n \in \mathbb{N}\} \subset c_0,$$

one has

$$\text{rad}_{\ell_\infty}(A) = \frac{1}{2}, \quad \text{rad}_{c_0}(A) = 1.$$

The missing centre is the constant sequence

$$(1/2, 1/2, \dots),$$

which belongs to ℓ_∞ but not to c_0 .

Short Version

This is a correct and useful counterexample to a possible extension from finite subsets to closed bounded subsets. It shows that the finite behaviour is not enough to determine the closed-bounded behaviour.

It does not provide a full characterisation of those pairs or subspaces for which the radius equality holds for all closed bounded subsets.

Possible Follow-Up

It may be worth checking with the authors whether they were already aware of this example, and whether they consider it interesting enough to include as a basic obstruction to the closed-bounded version of the problem.

Review Outcome

Archive this packet as correct, with the caveat that its contribution is best described as a natural counterexample or obstruction, not as a full solution of the characterisation problem.